# If the City Could Speak…

## *What Indicators miss and AI might find*

*Niklas Effenberger*
*Smart Urban Environments, Department of Urban Systems Engineering*
*Fraunhofer Institute for Industrial Engineering IAO*
*Nobelstrasse 12 | 70569 Stuttgart*

Fraunhofer IAO

## Introduction and Motivation

Urban systems are complex, dynamic and heterogeneous. They include both material and immaterial components, including infrastructure, housing, ecological systems, governance structures and socio-cultural dyna-mics (*see fig. 1)*.

**To evaluate and govern these systems, cities rely on indicators to:**

- Enable comparability across space and time
- Support monitoring, accountability and benchmarking
- Translate policy goals into operational targets

**However, these tools face conceptual and practical limitations** *(see tab. 1)*:

- They struggle to represent intangible dimensions like community cohesion, cultural life, ecological interde-pendencies, well-being, trust, …
- Quantitative indicators tend to privilege tangible, measurable and market-aligned outputs, while under-valuing non-monetary or systemic qualities.

## >> Central Research Problem <<

**How can qualitative, intangible factors be further strengthened in urban systems research and design?**

## Research Questions

1. **Do current indicator systems capture complex, context-specific urban dimensions adequately?**

2. **Can AI and emerging technologies create evaluation models that move beyond static, indicator-driven systems?**

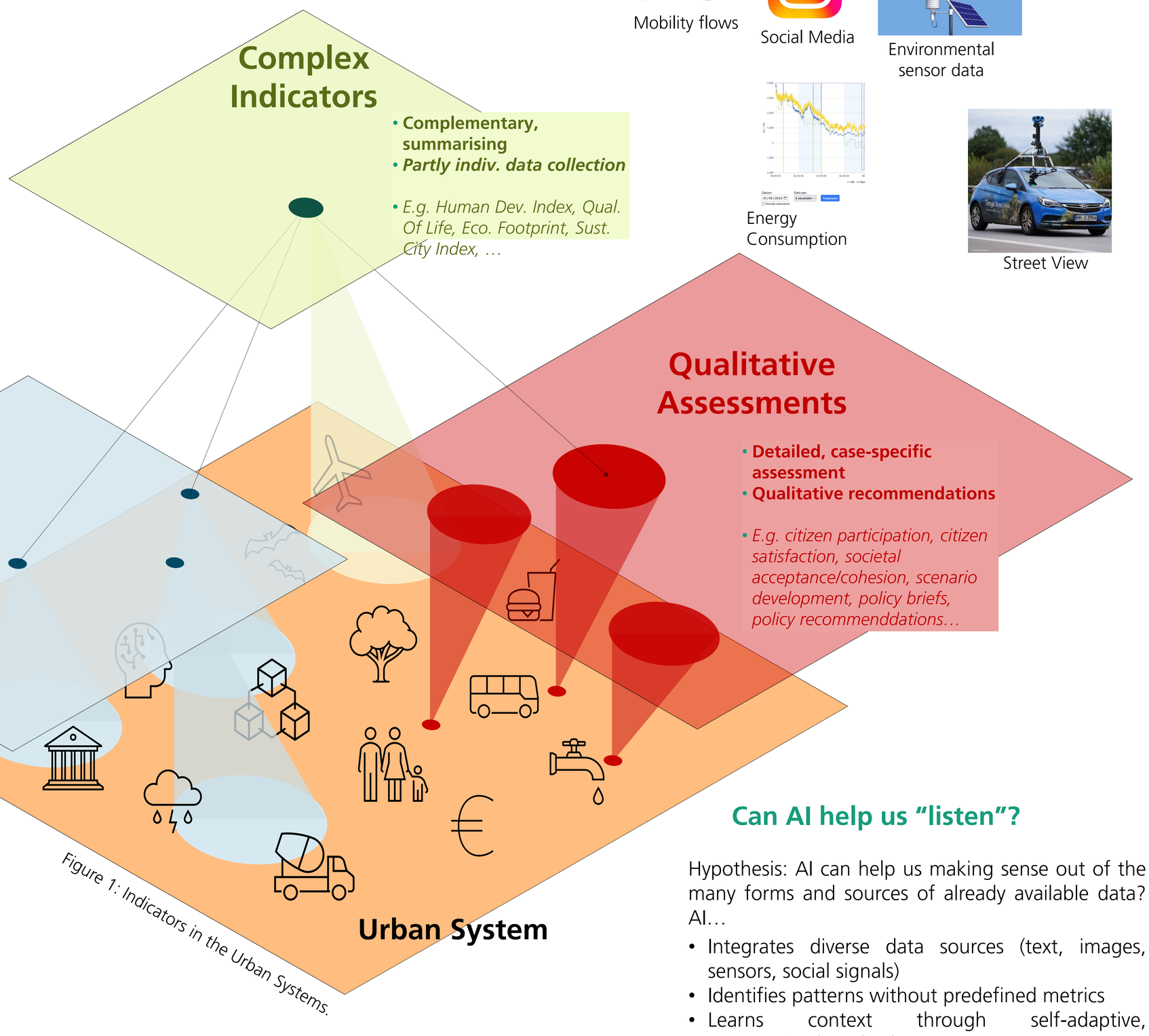



Figure 1: Indicators in the Urban Systems.

## The Indicators Gap

Table 1: Types of Indicators in Urban Assessment

| Type | Examples | Strenghts | Weaknesses |
|---|---|---|---|
| Economic | GDP, GMP | Comparable, trusted | Difficulties capturing intangibles |
| Complex | HDI, SDGs | Multi-dimensional | Hard to constitute and use; fragmented |
| Qualitative | Interviews, Policy Briefs | Context-rich | Hard to scale and generalise |

>> **Key Problem: Indicators are either over-simplified or complex, neither captures urban life fully.**

>> **Complex indicators cannot be site-specific and widely comparable at the same time.**

**This can lead to proliferation in complexity and number of indicators** *(see fig. 2).*

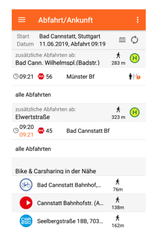
## >> The Indicator Spiral




Figure 2: The Indicator Adaptation Cycle

## The Urban Language Hypothesis

Do cities speak? They already emit a lot of publicly available data….



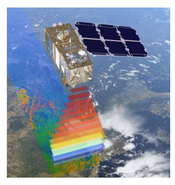

Mobility flows




News Outlets




Traffic Info



Satellite Obs

Social Media



Environmental sensor data



Energy Consumption



Street View

## Can AI help us "listen"?

Hypothesis: AI can help us making sense out of the many forms and sources of already available data? AI…

- Integrates diverse data sources (text, images, sensors, social signals)
- Identifies patterns without predefined metrics
- Learns context through self-adaptive, unsupervised methods
- Generates narrative insights further to numeric outputs
- Enables continuous, reflexive assessment of urban systems (…?)

## Approach

- Review of existing critique and the dominance of quantitative indicator assessments in policy
- Discourse Analysis on Indicators, metrics their persis-tence and the rationale behind the quantification approach
- Exploration of future visions & developping a concept-ual framework for AI-assisted urban assessment methods
- Pilot AI-supported narrative tools

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